

Batch 1 (Questions 1–20)

Question 1

Which characteristic of the Transformer architecture most directly enables efficient training on modern GPUs?

- A. Sequential hidden state updates
 - B. Parallel processing of all input tokens
 - C. Recursive weight sharing
 - D. Convolutional feature extraction
-

Question 2

Which component of the attention mechanism represents the information a token is searching for?

- A. Key
 - B. Value
 - C. Query
 - D. Embedding
-

Question 3

A company wants an AI model that excels at document classification and sentiment analysis rather than text generation. Which pre-training objective is most appropriate?

- A. Causal Language Modeling
 - B. Masked Language Modeling
 - C. Reinforcement Learning from Human Feedback
 - D. Diffusion Training
-

Question 4

Which statement best describes an embedding?

- A. A compressed vector representation capturing semantic meaning
 - B. A collection of training labels
 - C. A database index
 - D. A probability distribution over vocabulary
-

Question 5

Why have Transformers largely replaced RNNs for natural language processing?

- A. They require fewer parameters.
 - B. They eliminate the need for training.
 - C. They better capture long-range dependencies while allowing parallel computation.
 - D. They only work on fixed-length sequences.
-

Question 6

Which statement about self-attention is TRUE?

- A. Tokens only attend to adjacent tokens.
 - B. Every token can evaluate every other token in the same sequence.
 - C. Attention requires convolutional filters.
 - D. Self-attention only works during inference.
-

Question 7

Which mathematical operation is commonly used to calculate similarity between query and key vectors?

- A. Matrix inversion
 - B. Cross entropy
 - C. Dot product
 - D. Euclidean normalization
-

Question 8

What is the primary purpose of positional encoding in Transformers?

- A. Compress embeddings
 - B. Reduce training time
 - C. Provide information about token order
 - D. Remove duplicate tokens
-

Question 9

Which architecture is primarily associated with GPT models?

- A. Encoder-only

- B. Decoder-only
 - C. Encoder-Decoder CNN
 - D. Recurrent Encoder
-

Question 10

A training dataset contains billions of unlabeled web pages. Which learning paradigm makes best use of this data?

- A. Supervised Learning
 - B. Reinforcement Learning
 - C. Self-supervised Learning
 - D. Semi-supervised Learning
-

Question 11

Which of the following best explains latent space?

- A. Storage location for datasets
 - B. Low-dimensional representation where embeddings reside
 - C. Training cache
 - D. Weight initialization matrix
-

Question 12

Which optimization algorithm is widely used for training modern deep learning models?

- A. K-Means
 - B. Adam
 - C. Apriori
 - D. PageRank
-

Question 13

Which issue commonly affects deep RNNs during backpropagation?

- A. Data poisoning
- B. Gradient explosion or vanishing
- C. Hallucination

D. Token overflow

Question 14

In the attention mechanism, the Value vector represents:

- A. Information being requested
 - B. Information advertised
 - C. Information actually transferred
 - D. Positional information
-

Question 15

Which statement best differentiates model architecture from service architecture?

- A. Model architecture focuses on APIs.
 - B. Service architecture focuses on neural network layers.
 - C. Model architecture concerns internal AI computation, while service architecture concerns delivery and scalability.
 - D. There is no difference.
-

Question 16

A company wants maximum infrastructure control for deploying its own LLM. Which cloud service model is most appropriate?

- A. SaaS
 - B. PaaS
 - C. IaaS
 - D. FaaS
-

Question 17

Why is prompt caching more difficult for generative AI than traditional web caching?

- A. Models use SQL databases.
 - B. AI outputs may differ even for identical prompts.
 - C. GPUs cannot cache responses.
 - D. Prompts are encrypted.
-

Question 18

Which model architecture naturally supports next-token prediction?

- A. CNN
 - B. Decoder-only Transformer
 - C. Autoencoder
 - D. Vision Transformer
-

Question 19

Why do modern foundation models require significantly less data for downstream tasks?

- A. They memorize every possible answer.
 - B. They have already learned useful representations during pre-training.
 - C. They automatically retrieve answers from the internet.
 - D. They eliminate the need for optimization.
-

Question 20

Which statement about multiple attention heads is MOST accurate?

- A. Every attention head is manually assigned a fixed linguistic role.
 - B. Each attention head learns different patterns during training.
 - C. Attention heads only improve inference speed.
 - D. Attention heads eliminate positional encoding.
-

Answer Key**Q Ans**

- 1 B
- 2 C
- 3 B
- 4 A
- 5 C
- 6 B

Q Ans

7 C

8 C

9 B

10 C

11 B

12 B

13 B

14 C

15 C

16 C

17 B

18 B

19 B

20 B

Explanations

Q1: Transformers process all tokens simultaneously, enabling efficient GPU parallelization and significantly faster training than sequential architectures.

Q2: The **Query** represents "What information am I looking for?" Keys advertise available information, while Values provide the actual content.

Q3: Masked Language Modeling (MLM) is ideal for understanding tasks because it leverages bidirectional context, unlike CLM, which is optimized for generation.

Q4: Embeddings are dense vector representations that place semantically similar concepts close together in latent space.

Q5: Self-attention enables direct interaction between distant tokens without sequential memory bottlenecks, improving long-range dependency modeling.

Q6: In self-attention, every token can attend to every other token in the same sequence, allowing rich contextual understanding.

Q7: Query-key similarity is computed using the **dot product**, producing attention scores that determine relevance.

Q8: Because self-attention is position-agnostic, positional encoding provides information about token order.

Q9: GPT uses a **decoder-only Transformer** trained with causal language modeling for autoregressive text generation.

Q10: Self-supervised learning creates labels from unlabeled data, making it ideal for massive text corpora.

Q11: The latent space is the lower-dimensional space in which embedding vectors capture meaningful patterns.

Q12: Adam is the most widely used optimizer for training modern deep neural networks due to its adaptive learning rates.

Q13: Deep RNNs commonly suffer from **vanishing** and **exploding gradients**, making long-range learning difficult.

Q14: Values contain the actual information transferred after attention weights determine relevance.

Q15: Model architecture defines how the AI model works internally; service architecture focuses on deployment, APIs, scalability, reliability, and monitoring.

Q16: **IaaS** provides maximum control over compute, networking, and storage, making it suitable for organizations deploying custom AI infrastructure.

Q17: Generative AI is stochastic, meaning identical prompts may produce different outputs, complicating response caching.

Q18: Decoder-only Transformers predict one token at a time, making them ideal for autoregressive language generation.

Q19: Foundation models learn broad linguistic and semantic representations during pre-training, reducing downstream data and training requirements.

Q20: Attention heads naturally specialize during training, often focusing on grammar, semantic relationships, or coreference without explicit programming.

Batch 2 (Questions 21–40)

Question 21

A transformer is processing the sentence:

"The scientist presented her findings because she had completed the experiment."

Which mechanism primarily enables the model to determine that **"she"** refers to **"the scientist"**?

- A. Positional Encoding
 - B. Self-Attention
 - C. Tokenization
 - D. Gradient Descent
-

Question 22

Which statement best explains why Transformers outperform traditional RNNs on long documents?

- A. They eliminate embeddings.
 - B. They directly model relationships between distant tokens.
 - C. They use fewer trainable parameters.
 - D. They compress every sentence into a single hidden state.
-

Question 23

A language model is trained by predicting the next token while preventing access to future tokens.

Which training objective is being used?

- A. Masked Language Modeling
 - B. Span Corruption
 - C. Causal Language Modeling
 - D. Reinforcement Learning
-

Question 24

An organization wants an AI model capable of understanding entire documents before answering classification questions.

Which architecture is MOST appropriate?

- A. Decoder-only Transformer
- B. Encoder-only Transformer

C. CNN

D. RNN

Question 25

What is the primary purpose of Query vectors?

A. Store contextual information

B. Request relevant information from other tokens

C. Represent token positions

D. Reduce vocabulary size

Question 26

Which statement about attention heads is TRUE?

A. All heads learn identical linguistic relationships.

B. Each attention head learns different contextual patterns.

C. Attention heads replace embeddings.

D. Only one attention head is active during inference.

Question 27

Which problem is solved by introducing positional encoding?

A. Overfitting

B. Attention mechanisms have no inherent understanding of sequence order.

C. Vocabulary explosion.

D. Hallucination.

Question 28

A model predicts missing words located anywhere within a sentence using both left and right context.

This is characteristic of:

A. GPT

B. LLaMA

C. BERT

D. Claude

Question 29

Which statement BEST describes a Foundation Model?

- A. A model designed for a single business application.
 - B. A general-purpose model adaptable to many downstream tasks.
 - C. A database optimized for AI workloads.
 - D. A reinforcement learning agent.
-

Question 30

Why is Maximum Likelihood Estimation (MLE) fundamental to generative AI training?

- A. It minimizes GPU usage.
 - B. It encourages the model to assign higher probability to observed training data.
 - C. It prevents overfitting entirely.
 - D. It removes hallucinations.
-

Question 31

Which approach helps reduce memorization while improving generalization?

- A. Increasing learning rate indefinitely.
 - B. Eliminating dropout.
 - C. Regularization techniques.
 - D. Using smaller vocabularies.
-

Question 32

Which training phase is MOST associated with learning grammatical structure?

- A. Initialization
 - B. Early Training
 - C. Middle Training
 - D. Random Inference
-

Question 33

An AI engineer states:

"The model learned arithmetic, reasoning, and coding despite never being explicitly programmed."

This phenomenon is called:

- A. Gradient Descent
 - B. Transfer Learning
 - C. Emergent Capabilities
 - D. Overfitting
-

Question 34

Which statement BEST describes embeddings?

- A. They are encrypted versions of documents.
 - B. They convert semantic meaning into numerical vectors.
 - C. They are compressed databases.
 - D. They are identical to latent variables.
-

Question 35

Which retrieval technique relies primarily on semantic similarity?

- A. BM25
 - B. TF-IDF
 - C. Dense Retrieval
 - D. Keyword Search
-

Question 36

Why might a Retrieval-Augmented Generation (RAG) system produce incorrect answers even when retrieval succeeds?

- A. Retrieved context may not be incorporated effectively by the LLM.
 - B. Vector databases cannot store embeddings.
 - C. Retrieval automatically modifies model weights.
 - D. Retrieval disables attention.
-

Question 37

What distinguishes Knowledge-Augmented Generation (KAG) from RAG?

- A. KAG relies only on larger LLMs.
 - B. KAG incorporates structured knowledge representations.
 - C. KAG removes retrieval completely.
 - D. KAG replaces embeddings with SQL.
-

Question 38

A financial institution needs complete control over sensitive AI workloads while satisfying strict regulatory requirements.

Which deployment option is MOST appropriate?

- A. Public Cloud
 - B. SaaS
 - C. Private Cloud
 - D. Community Cloud
-

Question 39

Which cloud deployment principle recommends placing sensitive data close to where it is processed?

- A. Elasticity
 - B. Data Gravity
 - C. Fault Tolerance
 - D. Auto Scaling
-

Question 40

Which statement BEST differentiates Model Architecture from Service Architecture?

- A. Model Architecture concerns deployment while Service Architecture concerns optimization.
 - B. Service Architecture focuses on scalability, APIs, and availability, whereas Model Architecture focuses on internal AI computation.
 - C. Both terms describe exactly the same concept.
 - D. Service Architecture only applies to cloud environments.
-

Answer Key

Q Ans

21 B

22 B

23 C

24 B

25 B

26 B

27 B

28 C

29 B

30 B

31 C

32 C

33 C

34 B

35 C

36 A

37 B

38 C

39 B

40 B

Explanations

Q21

Correct Answer: B — Self-Attention

Self-attention allows the token "**she**" to attend directly to "**scientist**", enabling coreference resolution across the sentence.

Q22

Correct Answer: B

Unlike RNNs, Transformers create direct token-to-token interactions, avoiding sequential bottlenecks and improving long-range dependency modeling.

Q23

Correct Answer: C

Causal Language Modeling predicts the **next token** while masking future tokens, making it ideal for autoregressive generation.

Q24

Correct Answer: B

Encoder-only models (such as BERT) can use context from both directions, making them well suited for classification and document understanding.

Q25

Correct Answer: B

Queries ask: "**What information am I looking for?**"

Q26

Correct Answer: B

Attention heads naturally specialize during training, often learning different linguistic or semantic relationships.

Q27

Correct Answer: B

Attention itself is position-agnostic. Positional encoding injects sequence order into the model.

Q28

Correct Answer: C

BERT uses **Masked Language Modeling**, allowing it to use both preceding and following context.

Q29

Correct Answer: B

Foundation models are pretrained on massive datasets and can be adapted to many downstream tasks through prompting or fine-tuning.

Q30

Correct Answer: B

MLE trains the model to assign higher probability to observed training data, forming the basis of modern generative AI.

Q31

Correct Answer: C

Regularization techniques (dropout, weight decay, early stopping) encourage models to learn generalizable patterns instead of memorizing data.

Q32

Correct Answer: C

The middle stage of pretraining is where grammar, syntax, and topical coherence become established.

Q33

Correct Answer: C

Large models often exhibit **emergent capabilities**, developing reasoning and coding skills simply through scale.

Q34

Correct Answer: B

Embeddings convert words, sentences, or documents into dense numerical vectors that preserve semantic similarity.

Q35

Correct Answer: C

Dense retrieval uses neural embeddings to retrieve semantically similar content, unlike BM25 or TF-IDF, which rely on lexical matching.

Q36

Correct Answer: A

Even with successful retrieval, an LLM may ignore, misinterpret, or incorrectly integrate the retrieved context, resulting in inaccurate answers.

Q37

Correct Answer: B

KAG extends RAG by leveraging structured knowledge sources such as knowledge graphs, enabling more robust reasoning.

Q38

Correct Answer: C

Private cloud deployments offer the highest level of control for sensitive workloads and regulatory compliance.

Q39

Correct Answer: B

Data Gravity recommends keeping data close to where it is processed to reduce latency, improve security, and minimize transfer costs.

Q40

Correct Answer: B

Model Architecture focuses on how an AI model computes internally (e.g., transformers, attention), whereas Service Architecture addresses deployment concerns such as APIs, scalability, availability, monitoring, and reliability.

Question 41

A healthcare organization is developing an AI assistant that must generate clinical summaries while ensuring responses are based only on the latest approved medical guidelines. Which solution is MOST appropriate?

- A. Increase the model size
 - B. Fine-tune the model every day
 - C. Implement Retrieval-Augmented Generation (RAG)
 - D. Increase the temperature parameter
-

Question 42

A transformer model contains 48 attention heads.

What is the PRIMARY benefit of having multiple attention heads?

- A. Reducing vocabulary size
 - B. Allowing different semantic and syntactic relationships to be learned simultaneously
 - C. Eliminating positional encoding
 - D. Reducing the number of parameters
-

Question 43

A company is deciding between RAG and KAG.

Which business scenario BEST justifies selecting KAG?

- A. Retrieving recent news articles
 - B. Searching product manuals
 - C. Performing multi-hop reasoning across structured regulatory requirements
 - D. Summarizing meeting transcripts
-

Question 44

Why are embeddings preferred over one-hot encoding in modern language models?

- A. They require no training.
 - B. They preserve semantic similarity between concepts.
 - C. They eliminate tokenization.
 - D. They produce deterministic outputs.
-

Question 45

An engineer claims that attention alone understands word order.

Which response is MOST accurate?

- A. Correct—attention automatically captures sequence.
 - B. Incorrect—attention requires positional information.
 - C. Correct—but only during inference.
 - D. Correct—if enough layers are used.
-

Question 46

Which scenario BEST illustrates cross-attention?

- A. GPT predicting the next token
 - B. BERT filling in masked words
 - C. A translation model relating English input to French output
 - D. Self-attention inside one sentence
-

Question 47

During pretraining, a language model predicts:

"The capital of France is ____."

Which capability is primarily being learned?

- A. Database indexing
 - B. Statistical language and world knowledge
 - C. Rule-based programming
 - D. Reinforcement policies
-

Question 48

A model begins producing coherent grammar but still struggles with complex reasoning.

Which stage of pretraining is it MOST likely in?

- A. Random initialization
- B. Early training
- C. Middle training
- D. Post fine-tuning

Question 49

Which statement BEST explains why foundation models generalize well?

- A. They memorize every sentence.
 - B. They learn statistical patterns across massive datasets.
 - C. They use SQL databases.
 - D. They perform symbolic reasoning only.
-

Question 50

Why does dropout improve model robustness?

- A. It permanently removes neurons.
 - B. It randomly disables neurons during training, reducing over-reliance on specific pathways.
 - C. It reduces vocabulary.
 - D. It increases embedding dimensions.
-

Question 51

A bank wants to deploy an LLM internally due to strict privacy regulations while occasionally using public cloud GPUs during peak demand.

Which deployment strategy BEST fits?

- A. SaaS
 - B. Public Cloud
 - C. Hybrid Cloud
 - D. Community Cloud
-

Question 52

An auditor asks:

"Which cloud deployment principle recommends placing workloads in environments where they perform most efficiently?"

- A. Elastic Computing
- B. Workload Optimization
- C. Zero Trust
- D. Burst Scaling

Question 53

Which statement about Dense Retrieval is TRUE?

- A. It depends entirely on keyword matching.
 - B. It retrieves documents using semantic vector similarity.
 - C. It cannot retrieve unseen concepts.
 - D. It replaces embeddings with SQL indexes.
-

Question 54

A user searches:

"How do I terminate an employee?"

The retrieved document discusses **employee offboarding**.

Why would Dense Retrieval succeed where BM25 might fail?

- A. Dense Retrieval ignores meaning.
 - B. Dense Retrieval captures semantic similarity.
 - C. BM25 uses transformers.
 - D. Dense Retrieval removes embeddings.
-

Question 55

A transformer processes every token simultaneously.

Which hardware characteristic does this design best exploit?

- A. CPU cache
 - B. GPU parallel computation
 - C. Magnetic storage
 - D. SSD throughput
-

Question 56

Why is RLHF applied AFTER pretraining?

- A. RLHF learns grammar.
- B. RLHF aligns model behavior with human preferences.
- C. RLHF creates embeddings.

D. RLHF performs tokenization.

Question 57

Which statement BEST describes hallucination?

- A. Retrieval failure
 - B. Generation of plausible but incorrect information
 - C. Failure during optimization
 - D. Embedding corruption
-

Question 58

An AI system must answer questions using policies updated every hour.

Which approach minimizes stale knowledge?

- A. Larger parameter count
 - B. Daily fine-tuning
 - C. Retrieval-Augmented Generation
 - D. Lower temperature
-

Question 59

A sentence contains the word "bass."

How does attention help determine whether it refers to a fish or a musical instrument?

- A. By looking only at previous words
 - B. By evaluating contextual relationships with surrounding tokens
 - C. By checking a dictionary
 - D. By using reinforcement learning
-

Question 60

Which statement BEST summarizes the difference between pretraining and fine-tuning?

- A. Pretraining learns general knowledge; fine-tuning specializes the model for specific tasks.
- B. Fine-tuning replaces pretraining.
- C. Pretraining requires labeled data.
- D. Fine-tuning always changes the model architecture.

Answer Key

Q Ans

41 C

42 B

43 C

44 B

45 B

46 C

47 B

48 C

49 B

50 B

51 C

52 B

53 B

54 B

55 B

56 B

57 B

58 C

59 B

60 A

Explanations

Q41 — C

RAG grounds responses in the latest external knowledge, making it ideal for domains like healthcare where guidance changes frequently.

Q42 — B

Multiple attention heads learn different linguistic and semantic relationships simultaneously, enriching the model's representation.

Q43 — C

KAG excels when structured knowledge and multi-hop reasoning (e.g., regulatory rules or knowledge graphs) are required.

Q44 — B

Embeddings place semantically related words close together in vector space, unlike one-hot encoding, which treats all tokens as equally distinct.

Q45 — B

Attention is inherently position-agnostic; positional encodings provide sequence-order information.

Q46 — C

Cross-attention relates two different sequences, such as source and target languages in machine translation.

Q47 — B

Next-token prediction teaches both language structure and factual/world knowledge through statistical learning.

Q48 — C

The middle stage of pretraining is when grammar, syntax, and coherent language understanding emerge, before advanced reasoning capabilities.

Q49 — B

Foundation models generalize because they learn broad statistical patterns from massive, diverse datasets rather than memorizing individual examples.

Q50 — B

Dropout randomly disables neurons during training, encouraging redundancy and reducing overfitting.

Q51 — C

Hybrid cloud allows sensitive workloads to remain on private infrastructure while leveraging public cloud resources for burst capacity.

Q52 — B

Workload Optimization recommends placing workloads where they best balance performance, cost, compliance, and operational needs.

Q53 — B

Dense retrieval compares vector embeddings, enabling semantic matching even when exact keywords differ.

Q54 — B

Semantic embeddings recognize that "terminate an employee" and "employee offboarding" express related concepts despite different wording.

Q55 — B

Transformers are designed to exploit GPU parallelism by processing tokens concurrently.

Q56 — B

RLHF aligns pretrained models with human preferences for helpfulness, safety, and relevance.

Q57 — B

A hallucination occurs when a model confidently generates incorrect or unsupported information.

Q58 — C

RAG retrieves up-to-date documents at inference time, avoiding reliance solely on potentially outdated model parameters.

Q59 — B

Attention evaluates surrounding context (e.g., "guitar" vs. "river") to disambiguate the meaning of polysemous words like "bass."

Q60 — A

Pretraining builds broad language and world knowledge, while fine-tuning adapts the model for specific tasks or domains.

Batch 4 (Questions 61–80)

Question 61

An organization is building a chatbot for its legal department. The chatbot must answer questions using thousands of legal documents that change every week, while minimizing retraining costs.

Which approach is MOST appropriate?

- A. Increase model parameters to 500B
 - B. Fine-tune the model weekly
 - C. Implement Retrieval-Augmented Generation (RAG)
 - D. Increase the context window only
-

Question 62

A vector database stores document embeddings.

What is its PRIMARY purpose?

- A. Compress neural network weights
 - B. Perform semantic similarity search efficiently
 - C. Replace the tokenizer
 - D. Store training labels
-

Question 63

Which statement BEST differentiates sparse retrieval from dense retrieval?

- A. Sparse retrieval understands semantic meaning better.
 - B. Dense retrieval relies primarily on exact keyword matching.
 - C. Dense retrieval uses learned vector embeddings, while sparse retrieval relies on lexical matching.
 - D. They always produce identical search results.
-

Question 64

Which characteristic makes Knowledge-Augmented Generation (KAG) particularly valuable in regulated industries?

- A. Lower GPU utilization
- B. Structured knowledge supports explainable reasoning.
- C. Smaller embedding dimensions.
- D. Elimination of retrieval.

Question 65

A manufacturing company wants an AI system capable of reasoning across equipment hierarchies, maintenance history, and engineering relationships.

Which architecture is MOST suitable?

- A. Traditional RAG
 - B. KAG using a knowledge graph
 - C. BM25
 - D. TF-IDF
-

Question 66

Which statement BEST describes diffusion models?

- A. They directly generate an image in one step.
 - B. They iteratively remove noise from random noise to create realistic samples.
 - C. They classify images.
 - D. They predict the next word.
-

Question 67

In a diffusion model, what occurs during the reverse diffusion process?

- A. Noise is gradually removed to reconstruct meaningful data.
 - B. Training labels are generated.
 - C. Images are compressed into embeddings.
 - D. Tokens are predicted sequentially.
-

Question 68

What is the PRIMARY advantage of diffusion models over traditional GANs?

- A. They always train faster.
 - B. They generally produce more stable training and higher-quality image generation.
 - C. They eliminate latent spaces.
 - D. They require no training data.
-

Question 69

Which statement BEST describes the role of latent space?

- A. It stores database indexes.
 - B. It is the compressed feature space from which meaningful representations can be generated.
 - C. It stores model checkpoints.
 - D. It contains attention weights.
-

Question 70

An auditor reviews an AI system using embeddings for customer search.

What should the auditor primarily verify?

- A. GPU temperature
 - B. Whether semantically similar records produce nearby embeddings.
 - C. Tokenizer speed.
 - D. Screen resolution.
-

Question 71

An organization stores confidential customer embeddings inside a vector database.

Which security concern should receive the HIGHEST priority?

- A. Printer configuration
 - B. Unauthorized access to embeddings that could reveal sensitive information.
 - C. Browser cache.
 - D. Monitor brightness.
-

Question 72

Which deployment model provides the GREATEST operational responsibility while also providing the greatest flexibility?

- A. SaaS
 - B. PaaS
 - C. IaaS
 - D. FaaS
-

Question 73

A company experiences significant increases in AI inference demand during holiday shopping seasons.

Which hybrid cloud principle BEST addresses this requirement?

- A. Data Gravity
 - B. Burst Capacity
 - C. Knowledge Distillation
 - D. Fine-Tuning
-

Question 74

A disaster disables the primary AI inference environment.

Which hybrid cloud principle primarily supports continued AI service availability?

- A. Disaster Recovery
 - B. Prompt Engineering
 - C. Self-Attention
 - D. Fine-Tuning
-

Question 75

Which scenario BEST demonstrates data gravity?

- A. Moving all data daily between clouds regardless of cost.
 - B. Keeping sensitive medical records within the hospital while bringing computation to the data.
 - C. Replicating every document everywhere.
 - D. Compressing embeddings.
-

Question 76

A model predicts highly fluent text but occasionally invents citations.

Which mitigation is MOST appropriate?

- A. Increase randomness.
- B. Use Retrieval-Augmented Generation with trusted sources.
- C. Reduce embedding dimensions.
- D. Remove positional encoding.

Question 77

A chatbot frequently ignores retrieved context even though relevant documents are returned.

Which is the MOST likely explanation?

- A. Retrieval alone cannot guarantee the LLM properly incorporates retrieved information.
 - B. Embeddings are impossible to retrieve.
 - C. Attention has failed permanently.
 - D. The tokenizer removed documents.
-

Question 78

Why is PEFT (Parameter-Efficient Fine-Tuning) becoming increasingly popular?

- A. It always improves inference speed.
 - B. It adapts large models using relatively few trainable parameters.
 - C. It replaces pretraining.
 - D. It eliminates embeddings.
-

Question 79

An AI architect wants to customize an LLM for a specific insurance domain while minimizing GPU requirements.

Which approach is MOST appropriate?

- A. Retrain the foundation model from scratch.
 - B. Parameter-Efficient Fine-Tuning (PEFT).
 - C. Increase context length only.
 - D. Replace embeddings.
-

Question 80

Which statement BEST summarizes the relationship between pretraining, fine-tuning, and inference?

- A. Fine-tuning occurs before pretraining.
- B. Pretraining builds general capabilities, fine-tuning specializes the model, and inference applies the trained model to new inputs.
- C. Inference trains the model.
- D. Pretraining occurs after deployment.

Answer Key

Q Ans

61 C

62 B

63 C

64 B

65 B

66 B

67 A

68 B

69 B

70 B

71 B

72 C

73 B

74 A

75 B

76 B

77 A

78 B

79 B

80 B

Explanations

Q61

Correct Answer: C

RAG allows the chatbot to access frequently changing legal documents without repeatedly fine-tuning the underlying model.

Q62

Correct Answer: B

Vector databases are optimized for fast similarity searches using embeddings, enabling semantic retrieval.

Q63

Correct Answer: C

Sparse retrieval (e.g., BM25) depends on lexical overlap, whereas dense retrieval compares learned semantic embeddings.

Q64

Correct Answer: B

KAG leverages structured knowledge (such as knowledge graphs), enabling explainable, traceable reasoning—especially valuable in regulated industries.

Q65

Correct Answer: B

A knowledge graph captures relationships among equipment, maintenance events, and components, making KAG well suited for complex industrial reasoning.

Q66

Correct Answer: B

Diffusion models begin with random noise and progressively denoise it to generate realistic outputs.

Q67

Correct Answer: A

The reverse diffusion process iteratively removes learned noise to reconstruct a realistic sample.

Q68

Correct Answer: B

Compared with GANs, diffusion models generally exhibit more stable training and produce high-fidelity images, though they often require more inference steps.

Q69

Correct Answer: B

Latent space is a compressed representation that captures meaningful structure from which new samples or semantic relationships can be generated.

Q70

Correct Answer: B

Embedding quality should preserve semantic relationships so that similar customer records are represented by nearby vectors.

Q71

Correct Answer: B

Embeddings may encode sensitive information, making access controls, encryption, and governance critical.

Q72

Correct Answer: C

IaaS provides maximum flexibility and control, but also requires the organization to manage operating systems, networking, security, and infrastructure.

Q73

Correct Answer: B

Burst Capacity enables workloads to temporarily expand into public cloud resources during demand spikes.

Q74

Correct Answer: A

Hybrid disaster recovery enables workloads to fail over to alternate environments, improving resilience and business continuity.

Q75

Correct Answer: B

Data Gravity recommends keeping sensitive data where it resides and bringing computation closer to it rather than moving the data unnecessarily.

Q76

Correct Answer: B

Grounding responses with trusted retrieved documents reduces hallucinations and improves factual accuracy.

Q77

Correct Answer: A

Successful retrieval alone does not guarantee that the language model will correctly use or prioritize the retrieved context.

Q78

Correct Answer: B

PEFT methods (such as LoRA) adapt large models by updating only a small subset of parameters, significantly reducing compute and storage requirements.

Q79

Correct Answer: B

PEFT is well suited for domain adaptation when compute resources are limited, avoiding the cost of retraining an entire foundation model.

Q80

Correct Answer: B

Pretraining learns broad language and world knowledge, fine-tuning specializes the model for a domain or task, and inference is the application of the trained model to unseen inputs.

Batch 5 (Questions 81–100)

Question 81

A multinational bank wants an AI assistant that:

- answers using internal policies,
- cites authoritative sources,
- supports regulatory reasoning,
- explains how conclusions were reached.

Which architecture BEST satisfies these requirements?

- A. Standalone LLM
 - B. Retrieval-Augmented Generation (RAG)
 - C. Knowledge-Augmented Generation (KAG)
 - D. Larger Foundation Model
-

Question 82

Which statement BEST explains why decoder-only architectures dominate modern LLMs?

- A. They require fewer parameters than encoder models.
 - B. Their training objective aligns directly with autoregressive text generation.
 - C. They eliminate tokenization.
 - D. They don't require attention mechanisms.
-

Question 83

An engineer proposes removing positional encodings from a Transformer because "attention already understands sequence."

What is the MOST appropriate response?

- A. Agree—attention inherently captures order.
 - B. Agree—but only for long documents.
 - C. Disagree—attention is position-agnostic and requires positional information.
 - D. Disagree because embeddings already encode position.
-

Question 84

A model correctly predicts:

"The CEO announced quarterly earnings despite market uncertainty."

Which capability is MOST responsible for understanding that "earnings" relates to business rather than wages?

- A. Tokenization
 - B. Contextual attention
 - C. Dropout
 - D. Gradient clipping
-

Question 85

Which statement BEST explains why embeddings improve retrieval quality?

- A. They compress documents into ZIP files.
 - B. They capture semantic similarity beyond exact keywords.
 - C. They replace vector databases.
 - D. They eliminate preprocessing.
-

Question 86

A company fine-tunes a foundation model using only 0.5% of its parameters.

Which technique is MOST likely being used?

- A. CNN
 - B. PEFT (e.g., LoRA)
 - C. Gradient Boosting
 - D. BM25
-

Question 87

Which statement BEST distinguishes pretraining from RLHF?

- A. RLHF teaches grammar.
 - B. Pretraining learns statistical language patterns, while RLHF aligns outputs with human preferences.
 - C. RLHF replaces fine-tuning.
 - D. Both perform identical functions.
-

Question 88

A company stores millions of embeddings in a vector database.

Which search operation is MOST commonly performed?

- A. SQL JOIN
 - B. Nearest Neighbor Search
 - C. Binary Search
 - D. Merge Sort
-

Question 89

Why can dense retrieval outperform BM25 when users use different terminology from source documents?

- A. Dense retrieval ignores language.
 - B. Embeddings capture semantic meaning rather than exact lexical overlap.
 - C. BM25 uses GPUs.
 - D. Dense retrieval stores fewer documents.
-

Question 90

A regulator requires every AI-generated answer to reference authoritative policy documents.

Which solution BEST addresses this requirement?

- A. Increase model size.
 - B. Lower temperature.
 - C. Retrieval-Augmented Generation with citations.
 - D. Increase embedding dimensions.
-

Question 91

Which characteristic MOST differentiates KAG from traditional RAG?

- A. Larger context windows.
 - B. Structured knowledge enables reasoning across explicit relationships.
 - C. Smaller models.
 - D. Faster GPUs.
-

Question 92

An AI architect wants to reduce hallucinations WITHOUT modifying model weights.

Which approach is MOST appropriate?

- A. Larger vocabulary.
 - B. Retrieval-Augmented Generation.
 - C. More attention heads.
 - D. Additional embeddings.
-

Question 93

Which cloud deployment model generally offers the HIGHEST level of managed AI services?

- A. IaaS
 - B. Bare Metal
 - C. SaaS
 - D. Private Data Center
-

Question 94

A company wants maximum elasticity while maintaining sensitive data internally.

Which deployment architecture BEST satisfies this requirement?

- A. Public Cloud
 - B. Private Cloud
 - C. Hybrid Cloud
 - D. Standalone Data Center
-

Question 95

Which statement BEST explains why foundation models exhibit emergent capabilities?

- A. Engineers manually program new reasoning skills.
 - B. Scaling model size and data can produce capabilities not explicitly programmed.
 - C. Embeddings automatically generate reasoning.
 - D. RLHF creates reasoning from scratch.
-

Question 96

Which stage of training typically contributes MOST to acquiring broad world knowledge?

- A. Random initialization
 - B. Tokenization
 - C. Large-scale pretraining
 - D. RLHF
-

Question 97

A Transformer contains dozens of layers.

Why are multiple layers important?

- A. Each layer progressively builds richer representations from syntax to semantics.
 - B. Layers simply increase vocabulary.
 - C. Layers reduce embedding size.
 - D. Layers remove attention.
-

Question 98

A company deploys an AI assistant that retrieves policies from a vector database before generating answers.

Which component is MOST responsible for retrieving semantically similar documents?

- A. Tokenizer
 - B. Embedding model
 - C. RLHF
 - D. Optimizer
-

Question 99

Which statement BEST describes why modern AI systems combine foundation models with retrieval?

- A. Foundation models already know all future information.
 - B. Retrieval provides dynamic, current, and organization-specific knowledge unavailable during pretraining.
 - C. Retrieval replaces language models.
 - D. Retrieval improves GPU utilization.
-

Question 100

An AI architect must choose ONE statement that BEST summarizes modern enterprise generative AI.

Which is the BEST answer?

A. Bigger models eliminate the need for external knowledge.

B. Fine-tuning is always preferable to retrieval.

C. Enterprise AI combines pretrained foundation models, retrieval mechanisms, governance, and scalable cloud architectures to deliver accurate and trustworthy AI systems.

D. AI performance depends solely on parameter count.

Answer Key

Q	Ans
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81	C
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82	B
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83	C
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84	B
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85	B
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86	B
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87	B
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88	B
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89	B
----	---

90	C
----	---

91	B
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92	B
----	---

93	C
----	---

94	C
----	---

95	B
----	---

96	C
----	---

97	A
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98	B
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Q Ans

99 B

100 C

Explanations

Q81

Correct Answer: C

KAG is the best fit because it combines retrieval with structured knowledge (e.g., knowledge graphs), enabling explainable reasoning and traceable conclusions—important in regulated domains.

Q82

Correct Answer: B

Decoder-only architectures are trained using causal language modeling (next-token prediction), which aligns naturally with text generation during inference.

Q83

Correct Answer: C

Self-attention alone has no notion of token order. Positional encodings supply sequence information.

Q84

Correct Answer: B

Contextual attention allows the model to use surrounding words (e.g., "CEO," "quarterly") to infer that "earnings" refers to corporate financial results.

Q85

Correct Answer: B

Embeddings encode semantic relationships, allowing retrieval even when users and documents use different wording.

Q86

Correct Answer: B

PEFT techniques such as LoRA update only a small subset of parameters, enabling efficient domain adaptation.

Q87

Correct Answer: B

Pretraining builds language and world knowledge from large corpora; RLHF aligns model behavior with human preferences for helpfulness and safety.

Q88

Correct Answer: B

Vector databases commonly perform nearest neighbor (or approximate nearest neighbor) searches over embeddings.

Q89

Correct Answer: B

Dense retrieval leverages semantic similarity, whereas BM25 primarily relies on lexical overlap.

Q90

Correct Answer: C

RAG can retrieve authoritative documents and provide grounded responses with references, supporting regulatory requirements.

Q91

Correct Answer: B

KAG reasons over structured relationships, enabling more complex reasoning than traditional document retrieval.

Q92

Correct Answer: B

RAG grounds model outputs in external information without changing the model's weights.

Q93

Correct Answer: C

SaaS typically offers the highest level of abstraction and managed AI services, with the least operational responsibility for customers.

Q94

Correct Answer: C

Hybrid cloud balances elasticity (public cloud) with control and compliance (private infrastructure).

Q95

Correct Answer: B

Emergent capabilities arise as models scale in parameters, data, and compute, revealing behaviors not explicitly programmed.

Q96

Correct Answer: C

Large-scale pretraining is the primary phase where models acquire broad linguistic and world knowledge.

Q97

Correct Answer: A

Lower layers capture local syntactic patterns, while deeper layers progressively represent richer semantic and abstract concepts.

Q98

Correct Answer: B

The embedding model converts queries and documents into vectors that enable semantic retrieval.

Q99

Correct Answer: B

Foundation models provide general knowledge, while retrieval injects current, proprietary, or domain-specific information at inference time.

Q100

Correct Answer: C

Modern enterprise AI combines pretrained foundation models, retrieval, governance, security, and scalable deployment architectures to produce reliable and trustworthy systems.